Manav Patel

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PROFESSIONAL SUMMARY

Two-time autonomous systems champion – SUAS 2025 World Champions and IRoC-U 2026 National Champions – serving as Captain and Software Lead at Team Arrow ITNU. Specializes in autonomous UAV systems engineering: GPS-denied navigation, real-time computer vision models deployed on NVIDIA Jetson hardware, swarm coordination, and ROS-based mission-control architecture. Pursuing graduate research and industry opportunities in AI/ML and autonomous robotics.

ACHIEVEMENTS

- World Champions – Student Unmanned Aerial Systems 2025 (SUAS), USA (1st of ∼80 international university teams)
- National Champions – ISRO Robotics Challenge – URSC 2026 (IRoC-U) (1st of ∼590 registered teams nationwide) – Role: Captain
- Finalist – Robofest 5.0, Swarm Aerial Robotics Competition (Top 20 nationally)

EDUCATION

- **Nirma University – Institute of Technology, Ahmedabad** — Gujarat, India
  Bachelor of Technology - Computer Science and Engineering; CGPA: 9.12/10 — *Aug 2024 – May 2028*

TECHNICAL SKILLS

- **Languages:** C, C++, Python, Java, JavaScript, HTML, CSS, MATLAB, SQL (MySQL)
- **Frameworks & Libraries:** TensorFlow, PyTorch, OpenCV, YOLO, NumPy, Pandas, scikit-learn, CUDA, React.js, Node.js
- **Robotics & Simulation:** ROS (Robot Operating System), ArduPilot, PX4, MAVLink, Mission Planner, QGroundControl, Gazebo, SLAM
- **Tools & Platforms:** Git, GitHub, VS Code, Docker, Linux, Windows, Arduino, Raspberry Pi, NVIDIA Jetson, Simulink

EXPERIENCE

- **Team Arrow ITNU, SAE Nirma Collegiate Club** — Onsite

  Captain — *Jan 2026 – Present*
  - Captained a 13-member multidisciplinary team to the National Champions title at IRoC-U 2026, ISRO's national robotics competition, outperforming approximately 590 registered teams nationwide.
  - Directed end-to-end technical coordination across navigation, perception, and mission-control subsystems, unifying contributions from the full technical team into a single flight-ready autonomous system across 50+ autonomous flight test runs.

  Software Lead — *Jul 2025 – Present*
  - Architected the software stack for two concurrent autonomous UAV competition builds, defining system interfaces across navigation, real-time computer vision, and swarm-coordination modules.
  - Directed a team of 4 software engineers through regular code reviews and integration testing, building a predictive path-planning and obstacle-avoidance model alongside a computer-vision detection pipeline for coordinated swarm operations.

  Software Engineer — *Aug 2024 – Jul 2025*
  - Personally engineered the path-planning module and computer-vision detection pipeline for the SUAS 2025 competition UAV, integrating flight-control command logic with real-time object-detection inference and autonomous mission-execution logic.
  - Also developed and deployed the team's official website (teamarrow.in) using React.js and Node.js.

PROJECTS

- **Autonomous Survey UAV – SUAS 2025 World Champions** — [GitHub] [Team Arrow]
  Role: Software Engineer – Navigation & Computer Vision — *Aug 2024 – Jul 2025*
  - Engineered a custom coverage path-planning algorithm for autonomous airport-runway survey, achieving 15-acre area coverage per mission.
  - Fine-tuned a YOLO-based Foreign Object Debris (FOD) detection model on a self-annotated runway imagery dataset, achieving 91% precision at 45m UAV altitude, with automated geolocation and beacon deployment within 0.5m max positional error.
  - Developed a real-time mission-control GUI displaying live telemetry, autonomous status, and geo-referenced detection overlays for ground-operator monitoring.
- **Autonomous UAV Swarm – Robofest 5.0 Finalist** — [GitHub] [Team Arrow]
  Role: Software Lead – Swarm Coordination — *Aug 2025 – Mar 2026*
  - Formulated the inter-drone communication protocol and swarm-coordination logic for a 4-UAV formation system (1 lb each), achieving synchronized waypoint execution across the formation.
  - Implemented a forest-field survey and navigation algorithm combined with an RRT*-based predictive obstacle-avoidance model, achieving a 95% obstacle-avoidance success rate in simulation testing.
  - Fine-tuned a simulated landmine detection model achieving an 87% success rate, with automated safe-path re-routing for personnel traversal.
  - Integrated gesture- and voice-based flight control as an auxiliary manual-override interface.
- **Autonomous Agricultural Survey** — Team Arrow ITNU
  Role: Software Lead – Computer Vision — *Aug 2025 – Dec 2026*
  - Formulated a simple image-processing (IP) vision-based crop-health survey pipeline for an autonomous agricultural UAV, mapping GPS-tagged field zones to drive automated targeted irrigation and fertilization commands.
  - Developed a mission-control GUI for live monitoring of survey and irrigation status.
- **Mars Exploration UAV – IRoC-U 2026 National Champions** — [GitHub] [Team Arrow]
  Role: Captain – GPS-Denied Navigation & Docking — *Jan 2026 – Jul 2026*
  - Personally implemented an ORB-based visual-tracking and local-positioning pipeline for GPS-denied indoor navigation in a simulated Martian environment, achieving 20cm positional accuracy with zero external reference infrastructure.
  - Validated control-system dynamics and PID attitude loops in MATLAB/Simulink before deployment to flight-control firmware.
  - Engineered the autonomous self-charging docking sequence, achieving a 92% docking success rate.
  - Trained a YOLO-based model for Martian terrain feature identification, paired with a cosine-similarity feature-matching pipeline for landmark re-identification, achieving 87% overall accuracy.